

Lecture (1)

Introduction to Wireless Sensor Networks

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What you should get out of the course

❑ In depth understanding of:

- To understand the characteristics, basic concepts in Wireless sensor networks
- To understand Wireless Sensor Network Structure
- To understand the characteristics, basic concepts in to Internet of Things.
- IoT Networking
- Connectivity Technologies
- M2M Communication
- Programming with Arduino and Programming with Raspberry Pi
- Smart Homes, Smart Cities and Industrial IoT

Course Assessment

❑ Total Degree **150**

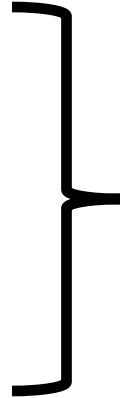
❑ Grading:

▪ Midterm Exam

▪ Project

▪ Assignments

▪ Participation



60 marks

▪ Final Exam

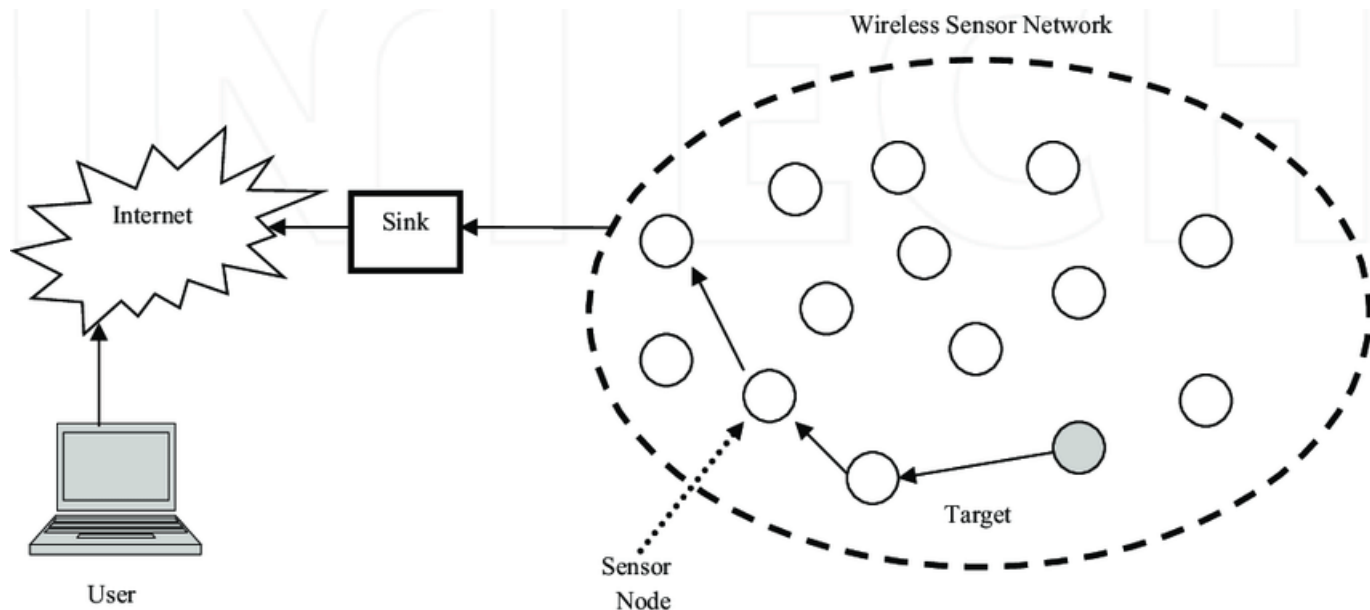
90 marks

What is the Wireless Sensor Network?

- ❑ **A Wireless Sensor Network (WSN):** is a distributed network and it comprises a large number of distributed, self-directed, tiny, low powered devices called sensor nodes.
- ❑ wireless network designed to monitor and measure various physical or environmental conditions. These include but are not limited to:
 - Temperature
 - Sound
 - Vibration
 - Pressure
 - Motion
 - Pollutants

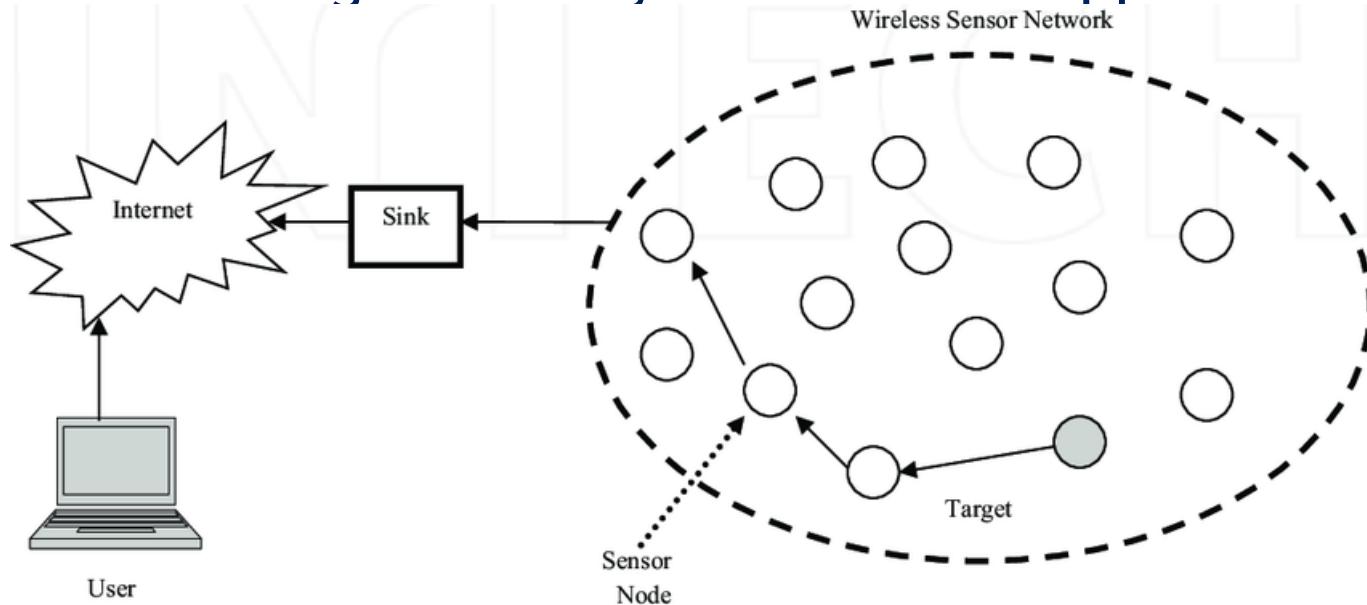
What is the Wireless Sensor Network?(Cont.)

- ❑ The data collected by the sensor nodes is transmitted cooperatively through the network to a central location, referred to as a **sink** or **base station**.
- ❑ At the sink, the gathered data can be observed and analyzed.



Role of the Sink or Base Station

- ❑ The **sink** acts as a bridge between users and the network.
- ❑ Users can interact with the network by injecting queries into the system and retrieving the results directly from the sink.
- ❑ This functionality makes WSNs highly efficient for real-time monitoring and analysis in diverse applications.



Components of a Wireless Sensor Network

- ❑ A typical WSN consists of **hundreds to thousands of sensor nodes** that are interconnected through radio communication (transmission of information through radio waves, a type of electromagnetic wave, without the need for **physical connections like wires or fiber optics**. It is widely used in various applications, including broadcasting, mobile communication, satellite systems, and military operations).
- ❑ Each node in the network is equipped with the following key components:
 - **Sensing Devices:** Measure the targeted physical or environmental parameters.
 - **Computing Devices:** Process and store the sensed data.
 - **Radio Transceivers:** Facilitate communication between nodes.
 - **Power Components:** Provide energy to operate the sensor nodes, often powered by batteries or energy-harvesting mechanisms.

Characteristics of wireless sensor networks

Characteristics of Wireless Sensor Networks (WSN)

- ❑ Wireless Sensor Networks (WSN) are defined by several key characteristics that heavily influence their software architecture. These characteristics are crucial for the effective operation and design of the network.

Fault Tolerance

- Each node in a WSN is prone to unexpected failure.
- **Fault tolerance** is the ability of the network to maintain its functionality even in the event of sensor node failures.
- Sensor nodes may fail due to power depletion, physical damage, or environmental interference, but their failure should not affect the overall task of the network.
- Protocols like **Automatic Repeat Request (ARQ)**, **Forward Error Correction (FEC)**, and **Hybrid ARQ** are used for fault detection and recovery.

Characteristics of wireless sensor networks (Cont.)

Mobility of Nodes

- Nodes in the network can move within the sensor field depending on the type of application. This movement can improve communication efficiency.

Dynamic Network Topology

- The connection between sensor nodes follows a standard topology, but WSNs must be capable of adapting to dynamic changes in topology.

Communication Failures

- If any node in the WSN fails to exchange data with other nodes, it should immediately inform the base station or gateway node to ensure timely data recovery.

Characteristics of wireless sensor networks (Cont.)

Heterogeneity of Nodes

- The sensor nodes deployed in the WSN may vary in type and capabilities, requiring them to cooperate effectively in the network.

Scalability

- WSNs can contain hundreds or even thousands of nodes. The network should be designed to scale efficiently with the increase in node count.

Independency

- WSNs should be capable of operating without a central control point, enabling independent functioning of the sensor nodes.

Programmability

- The network should support reprogramming or reconfiguring to adapt to dynamic changes in the environment or network conditions.

Characteristics of wireless sensor networks (Cont.)

Utilization of Sensors

- The sensors should be utilized in a manner that maximizes performance while minimizing energy consumption.

Self-Organization

- Given the large number of nodes, manual network organization by a user is impractical. The nodes must be able to self-organize and partition the network for efficient operation.
- The network should be robust and capable of tolerating node failures and variations in topology.

Concurrency and Cooperative Processing

- Sensor nodes direct data flow through the network and perform multiple tasks simultaneously, such as capturing, processing, and transmitting sensor data.
- **Cooperative processing** allows nodes to aggregate and preprocess data to reduce network traffic and improve efficiency. 11

Characteristics of wireless sensor networks (Cont.)

Energy Efficiency

- Energy efficiency is crucial in WSNs as nodes have no external power source. The energy consumption should be minimized by reducing communication between nodes.
- Protocols, OS power management, and task scheduling strategies play an essential role in optimizing energy usage.

Why use a WSN?

Wireless Sensor Networks (WSNs) offer numerous advantages that make them a compelling choice for a wide range of applications. Below are the key reasons:

- Wireless Communication:**

No need for complex infrastructure; WSNs enable seamless communication without physical connections.

- Plug and Play:**

Nodes can be deployed easily by simply placing them in the target area, reducing setup time.

- Cost-Effective:**

Built with inexpensive, off-the-shelf components, WSNs offer a low-cost alternative for data acquisition systems.

- High-Density Node Placement:**

WSNs support the deployment of a large number of nodes in a given area, enabling detailed and precise monitoring of physical or environmental conditions.

TinyOS

- ❑ **TinyOS** is an open-source, lightweight operating system specifically designed for wireless sensor networks (WSNs) and embedded systems.
- ❑ Developed at the University of California, Berkeley, it is optimized for resource-constrained devices with limited memory, processing power, and energy.
- ❑ **Advantages of TinyOS**
 - **Energy Efficiency:** Designed to minimize energy consumption in sensor nodes.
 - **Scalability:** Suitable for both small and large sensor networks.
 - **Flexibility:** Component-based architecture allows easy customization for various applications.
 - **Community Support:** Backed by a strong community and extensive documentation.

Applications of wireless sensor network

- ❑ **Military Applications** : Real-time monitoring for strategic decision-making.
- ❑ **Area Monitoring** : When sensors detect a specific event, they report it to the base station for appropriate action.
- ❑ **Transportation** : Collecting real-time traffic information to improve transportation models as well as Alerting drivers about congestion and traffic issues.
- ❑ **Health Applications**: Supporting interfaces for disabled patients, Integrated patient monitoring and diagnostics, and Drug administration in hospitals.
- ❑ **Environmental Sensing**: Monitoring natural phenomena such as volcanoes, oceans, glaciers, and forests.
- ❑ **Structural Monitoring**: Monitoring the movement within buildings and infrastructure, such as Bridges, Flyovers and Embankments.
- ❑ **Industrial Monitoring**: Developed for condition-based maintenance (CBM) of machinery and Offers significant cost savings and enhances operational efficiency.

Classifications of WSN

□ **The classification of Wireless Sensor Networks (WSN) is as follows:**

1. Static and Mobile Network
2. Deterministic and Nondeterministic Network
3. Single-Sink and Multi-Sink Network
4. Single-Hop and Multi-Hop Network
5. Self-Reconfigurable and Non-Self-Configurable Network
6. Homogeneous and Heterogeneous Network

Static and Mobile Network

□ Static and Mobile Network

- In many applications, all the sensor nodes are fixed and do not move. These networks are referred to as **static networks**.
- In certain applications, especially in biological systems, mobile sensor nodes are required. These are known as **mobile networks**.
- An example of a mobile network is **animal monitoring**.

Deterministic and Nondeterministic Network

□ Deterministic and Nondeterministic Network

- In a **deterministic WSN**, the position of each sensor node is calculated and fixed.
- The pre-planned deployment of sensor nodes is feasible in only a limited number of applications.
- If the position of sensor nodes cannot be determined due to factors like harsh environments or hostile operating conditions, the network is **nondeterministic**. These networks require a complex control system to operate.

Single-Sink and Multi-Sink Network

❑ Single-Sink Wireless Sensor Network (WSN)

- In a **Single-Sink WSN**, all sensor nodes send their data to a single central node, known as the sink or base station. The sink collects, processes, and forwards the data to external systems.

❑ Advantages:

1. **Simpler Network Design:** Easier to implement and manage due to a single data collection point.
2. **Cost-Effective:** Reduces infrastructure requirements.
3. **Centralized Processing:** All data aggregation occurs at one location, simplifying analytics.

❑ Disadvantages:

1. **Bottleneck Issues:** Heavy traffic around the sink can lead to network congestion.
2. **Scalability Challenges:** Limited ability to handle large-scale networks.
3. **Energy Depletion:** Nodes near the sink exhaust their energy faster due to frequent data transmission, leading to a **hotspot problem**.
4. **Single Point of Failure:** If the sink fails, the entire network becomes non-functional

Single-Sink and Multi-Sink Network

❑ Multi-Sink Wireless Sensor Network (WSN)

- In a **Multi-Sink WSN**, multiple sink nodes are deployed, and sensor nodes can send their data to the nearest sink. This setup distributes the workload and enhances network performance.

❑ Advantages:

1. **Improved Scalability:** Suitable for large-scale networks, as multiple sinks reduce congestion.
2. **Balanced Energy Consumption:** Traffic is distributed among multiple sinks, mitigating the hotspot problem.
3. **Increased Reliability:** If one sink fails, other sinks can continue functioning, ensuring network resilience.
4. **Reduced Latency:** Data travels shorter distances to the nearest sink, reducing delays.

❑ Disadvantages:

1. **Higher Cost:** Deployment and management of multiple sinks increase costs.
2. **Complexity in Design:** Requires more sophisticated routing protocols to coordinate among multiple sinks.
3. **Inter-Sink Communication Overhead:** Synchronization between sinks may introduce additional overhead.

Single-Hop and Multi-Hop WSN

□ Single-Hop and Multi-Hop WSN

- In a **single-hop WSN**, sensor nodes are directly connected to the base station.
- In a **multi-hop WSN**, peer nodes and cluster heads are used to relay the data, reducing energy consumption.

□ Self-Reconfigurable & Non-Self-Configurable WSN

- In most **self-reconfigurable WSNs**, sensor nodes are capable of organizing themselves and maintaining connections. They work collaboratively with other sensor nodes to accomplish tasks.
- In a **non-self-configurable WSN**, the sensor network cannot organize itself and cannot rely on a control unit to collect information.

Homogeneous and Heterogeneous WSN

□ Homogeneous and Heterogeneous WSN

- In a **homogeneous WSN**, all sensor nodes have similar energy consumption, computational power, and storage capabilities. This network follows a single architecture and operating system (e.g., **Ethernet LAN**).
- In a **heterogeneous WSN**, some sensor nodes possess higher computational power and energy than others.
- **Homogeneous sensor networks** have nodes with identical properties in terms of computation, communication, memory, energy level, and reliability.
- **Heterogeneous sensor networks** feature nodes with varying capabilities in computation, communication, memory, energy level, and reliability.

Wireless Sensor Networks (WSN) and the Internet of Things (IoT) are related but distinct concepts in networking and embedded systems.

Feature	Wireless Sensor Networks (WSN)	Internet of Things (IoT)
Definition	A network of spatially distributed sensors that monitor and collect data from the environment.	A broader ecosystem where devices, sensors, and systems communicate and interact over the internet.
Scope	Primarily focused on data collection and monitoring.	Encompasses data collection, processing, analytics, and automation.
Connectivity	Usually uses short-range wireless communication (Zigbee, Bluetooth, LoRa, etc.).	Uses various communication protocols, including Wi-Fi, 5G, LoRaWAN, and cloud-based services.
Architecture	Typically a centralized system with a base station aggregating data from multiple sensors.	A distributed and scalable system with cloud and edge computing capabilities.
Data Processing	Data is processed locally or at a gateway.	Data is processed locally (edge computing) or in the cloud for further analysis and decision-making.
Example Applications	Environmental monitoring, industrial automation, military surveillance.	Smart homes, healthcare, smart cities, autonomous vehicles, and more.

Any Question